

Exercise Session 12: Electrostatics Recap

Question 1 (From Section 1.3.4):

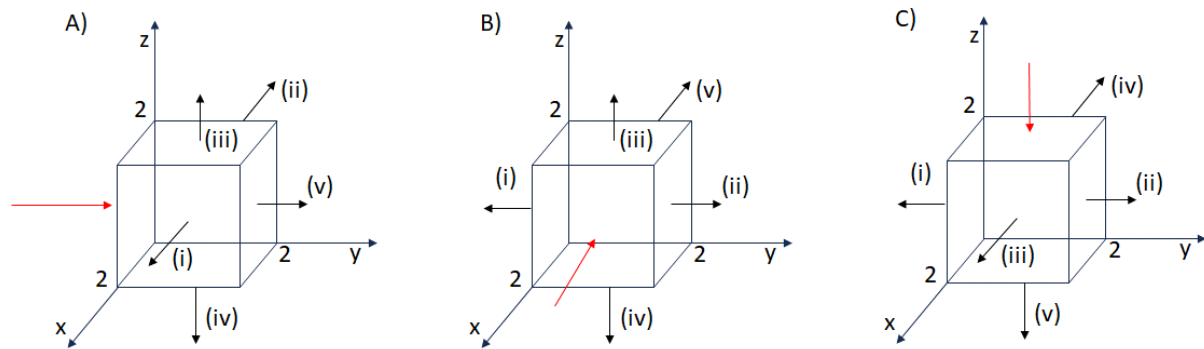
Answer the following:

- a) Divergence theorem is given as:

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}.$$

Explain both sides of the divergence theorem with words (*no math allowed*). What does the left-hand side represent physically? What does the right-hand side represent physically?

- b) Considering the integral calculus and more specifically the surface integrals, calculate the total flux of $\mathbf{f} = 2z\hat{x} + 3x\hat{y} + 5x^2yz\hat{z}$ over five sides for the three cases indicated in the figure below. Consider upward and outward as positive direction. Consider that the side of the red arrow is open.



Hint: Define first $d\mathbf{a}$. After finding $\int \mathbf{f} \cdot d\mathbf{a}$ for all surfaces the total flux is the sum of all.

Question 2 (From Section 2.2.3):

Consider an infinitely long hollow cylindrical tube with radius R , which carries a uniform surface charge σ . Use Gauss's law to find the $\mathbf{E}$ -field inside and outside the tube.

Hint: Introduce a Gaussian surface and identify what Q_{enc} is inside and outside the hollow tube.

Question 3 (From Section 2.2.3 and 2.3):

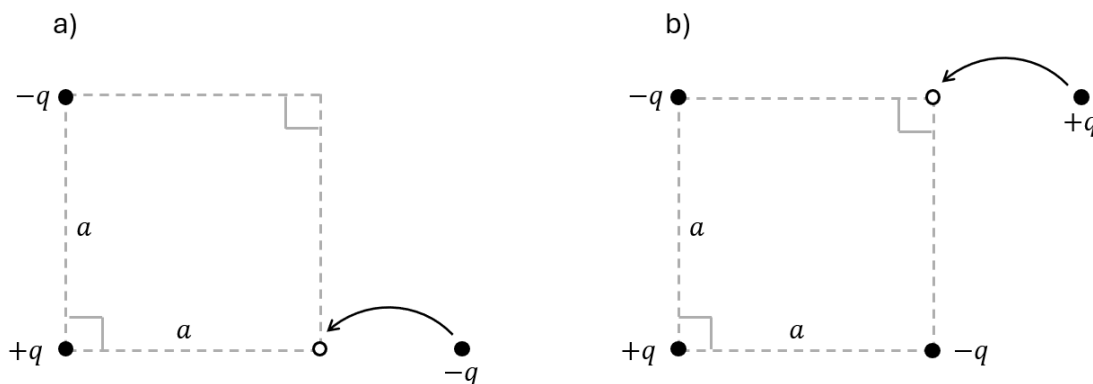
Consider a sphere with radius R carrying a *non-uniform* volume charge density $\rho(r) = kr$, with k being a constant.

- i) Find the electric field $\mathbf{E}$ inside and outside the sphere.
- ii) Find the potential V inside and outside the sphere.

Hint: Design your Gaussian surface appropriately for the inside/outside case and use Gauss's law. Remember that since the volume charge density is not uniform, Q_{enc} inside a Gaussian surface is now found by integrating it over the volume enclosed by the Gaussian surface.

Question 4 (From Section 2.4):

- a) Two opposite charges are placed at two corners of a square with side a as shown in figure a) below. How much work is required to bring a third charge, $-q$, from far away and place it in a third corner of the square as shown in the figure?
- b) After the charge in a) has been placed, what is the work required to bring a fourth charge, $+q$, from far away and place it in the fourth corner of the square as shown in figure b)?
- c) What is the work required to assemble the whole configuration of four charges?



Hint: Find first V . Consider the interaction of all charges with each other.

Question 5 (Section 4.3):

A thick spherical shell of linear dielectric material occupies the region

$$a < r < b,$$

with permittivity ϵ . A total free charge $+Q_f$ is placed at the center (a point charge at $r = 0$). Assume electrostatic equilibrium and spherical symmetry.

(a) When a dielectric is present, explain why applying Gauss's law to $\mathbf{E}$ compared to $\mathbf{D}$ is often difficult in highly symmetric systems (spherical, cylindrical, planar symmetry). And explain why using

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f\text{enc}}$$

is often simpler?

(b) Find $\mathbf{D}(r)$ in the three regions: (i) $0 < r < a$, (ii) $a < r < b$, (iii) $r > b$.

(c) Use $\mathbf{D} = \epsilon_0 \mathbf{E}$ in vacuum and $\mathbf{D} = \epsilon \mathbf{E}$ in the dielectric to find $\mathbf{E}(r)$ everywhere.

Question 6 (Section 4.3 and 4.4):

A conducting sphere of radius R contains a uniform charge of $Q^2/2$. It is surrounded by a thick rubber shell out to radius a , which is a linear dielectric of permittivity ϵ .

- a) Find the $\mathbf{E}$ -field
 - i. in the sphere.
 - ii. in the rubber.
 - iii. outside the rubber.
- b) Find the polarization in the system. What part of the system is polarized? Also find the charge in the rubber. What kind of charge is this?

Hints:

- a) *Make a sketch! Imagine what your system looks like and what information given above is useful for each calculation. How can you find the electric field in a simple way?*
- b) *What is the effect of the charged sphere on the rubber? What kind of charges do you know and when do each of them occur?*